Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/07/2006
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Project Title:	Performance and Sleep Consequences of Slam Shifts in Schedule		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Zip Code:	15213	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
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This project comprises a ground-based study of how human sleep, circadian rhythms and performance are affected by abrupt “slam shifts” as are encountered by astronauts on space station when the work-rest schedule needs to be changed (e.g., to accommodate a supply vehicle rendezvous). The study will additionally determine whether 90-minute naps in the workday following such shifts can be helpful. Comparisons will also be made with data from previous University of Pittsburgh studies (NAG9-1234) of repeated gradual phase shifts that are within NASA crew scheduling constraints (Appendix K).

Task Objective: The experiment seeks to answer several questions about coping with slam shifts (i.e., abrupt shifts in work/rest schedule of 6h), including the following: 1) In the light levels and comparative social isolation typical of ISS, how long does the human Endogenous Circadian Pacemaker (ECP) take (i.e. how many days) to entrain completely (at

Task Description:	the correct phase angle) to: a) a 6h phase advance (slam shift out)? and b) a 6h phase delay (slam shift back)? 2) To what degree, and for how many nights, is objectively and subjectively recorded sleep disrupted by: a) a 6h phase advance (slam shift out)? and b) a 6h phase delay (slam shift back)? 3) To what degree, and for how many days, is objectively measured performance and subjectively reported alertness and mood disrupted by a) a 6h phase advance (slam shift out)? and b) a 6h phase delay (slam shift back)? 4) To what extent (if any) are the disruptions noted in #2 and #3 ameliorated by a 90-minute daytime nap scheduled for the workday following each of the two slam shifts? 5) To what extent (if any) are the total disruptions in circadian rhythms, sleep, alertness and performance occasioned by slam shifts better or worse than those occasioned by the repeated gradual (appendix K - like) phase shifts we have studied in our previous grant (NAG 9-1234)? 6) How will the disruptions noted in #3 vary by the particular task used? Task Significance: Impact on America: 1. Industry Affiliates: None 2. Invovative Technologies Developed: None 3. Research Audience: Our primary research audience includes NASA crew schedulers and human factors managers. However, industrial safety and medical professionals concerned with workers who have to work at night or fly across time zones are also included in our research audience. 4. Graduate Student Employment: None 5. No. times Research presented on Magazine Cover: None
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are approximately 20 million Americans who are required to work outside of the "usual" 8am to 6pm daylight schedule. Millions more have to travel to different countries on business. Thus, every day, millions of U.S. night workers and air travelers are required to change the timing of their biological clock (circadian system) so that it is correctly aligned to their new work/rest schedule. This very often causes distress related to sleep disruption and to impairments of mood, alertness and performance while awake. Such distress may in some cases lead to accidents and illness. Very often such schedule changes are difficult to make abruptly, and often it is more reasonable to adopt a more gradual approach, trickling in the time changes little by little, rather than in one acute shift in schedule. We researched such gradual shifts in previously funded NASA work. Here, for comparison, we study the effects of abrupt shifts. We are also testing whether an afternoon nap on the first shift day can be an effective countermeasure, and whether different directions of shift change have different effects. While the present task is mostly concerned with work/rest scheduling in space, it also addresses an important question for all human beings trying to phase shift their schedule. Thus earth benefits will accrue to shift workers and air travelers from this task which will have a potentially positive impact on America. The knowledge about the response of humans to phase shifts derived from this research may lead to alternative strategies to ameliorate these problems.
Task Progress:	Task Progress: It is too early to present specific results from this study. A total of 54 subjects have been recruited and consented for the study. The following table presents details of their age, gender and race. Of these 54 subjects, 9 have passed through or are currently part way through our rigorous screening procedures which include detailed physical and mental health screening interviews and tests as well as a night spent in our sleep lab with detailed polysomnographic measures taken. Of those subjects, 7 (3 Female, 4 Male) have completed the entire 16-day protocol. One subject (f, 42y) started the protocol, but had to quit the protocol part way through when her brother died. SUPPLEMENTAL RESEARCH PROGRESS FOR FINAL REPORTING Technical content We accommodated the unexpected movement of the termination date of this grant by NASA from 6/1/2009 to 9/30/2006 (with associated cuts in funds) by running fewer subjects through the protocol, rather than by reducing the length or complexity of the protocol. In order to save money we also did not assay the urine samples we had already collected. Fortunately we had not yet run any subjects through the “nap" condition of the protocol, so we did not run any subjects through the “nap” protocol and were thus not able to test any hypotheses regarding naps as a countermeasure. The hypotheses regarding circadian rhythms will thus be tested using only the circadian rhythms in core body temperature and urine volume. Hypotheses regarding sleep, mood, alertness and performance will be tested using the measures outlined in the grant application. We ran 8 subjects through this protocol by 9/30/2006, simultaneously performing quality control on the data, archiving the data and performing summary statistics on the data. Data analysis has not yet been completed. Deliverables The final sample data set will be compared (with the obvious limitation in sample size resulting from the untimely truncation) to the other two data sets in this series of experiments (supported by NAG9-12334) which involved repeated 2h phase delays and repeated 0.5h phase advances. This may allow some limited conclusions to be made regarding the important question as to whether slam shifts or repeated gradual phase shifts are a better means of phase shifting the work/rest schedule of astronauts. If feasible given the reduced sample size, these conclusions will form the basis of a manuscript for peer review to a journal such as Aviation Space and Environmental Medicine (ASEM), or the Journal of Gravitational Physiology. This was not accompished before 9/30/2006. The 2h delay experiment was published in Monk TH, Buysse DJ, Billy BD, and DeGrazia JM: Using nine 2-h delays to achieve a 6-h advance disrupts sleep, alertness and circadian rhythm. Aviation, Space and Environmental Medicine,75:1049-1057, 2004. This paper acknowledges support from both NAG9-1234 and NNJ04HF76G. The remaining support of NNJ04HF76G was used to support the work needed to revise a companion paper to ASEM which describes the 0.5h phase advance experiment, as well as the neuroendocrine and performance data from both NAG9-1234 experiments. This manuscript was submitted to Aviation, Space and Environmental Medicine and was published in the 2006 issue of that journal (Monk TH, Buysse DJ, and Billy BD: Using daily 30-min phase advances to achieve a 6-h advance: Circadian rhythm, sleep and alertness. Aviation,

	Space and Environmental Medicine, 77:677-686, 2006).
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